

Hyperglycemia Symptoms



Diabetes and its complications

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ILO

- **At the end of the lecture, dental students should be able to:**
 - **Identify definition, classification (types) of diabetes**
 - **List risk factors for diabetes.**
 - **Discuss pathophysiology of type 1 and 2 diabetes.**
 - **Identify clinical presentation and complications of diabetes**
 - **Recognize how to diagnose diabetes**
 - **Describe lines of treatment of type 1, type 2 DM**
 - **How to do follow up diabetes.**
 - **Identify dental considerations in diabetes.**

Diabetes mellitus (DM)

- DM is a group metabolic disorder characterized by hyperglycemia with disturbance of carbohydrate, fat and protein metabolism due to absolute or relative insulin deficiency, resistance or both.
- In 2021, 537 million people had diabetes worldwide, with type 2 making up about 90% of all cases.
- Prevalence of the disease continues to increase, in developed countries (low-and middle-income nations).
- Rates are similar in women and men, with diabetes being the seventh leading cause of death globally.

Risk factors

- **Family history** (parent or sibling with type 2)
- **Overweight/ obese**
- **Inactivity.**
- **Age** (type 2 DM rises with age (45 year)).
- **Race and Genetics:**
 - Type 1 DM is more common in Caucasians and European countries.
 - Type 2 DM is common in African, Asians and Hispanics.
- **Hypertension, hyperlipidemia.**



Classification (types) of Diabetes

- **Diabetes may be primary (idiopathic) or secondary.**

Classification of DM		
Primary	Type 1	Insulin-dependent (IDDM) or juvenile-onset diabetes
	Type 2	Non-insulin-dependent (NIDDM) or maturity-onset diabetes
	Type 3	T3DM or Gestational diabetes
Secondary	• Drugs (corticosteroids, thiazide diuretics, beta-blockers). • Endocrine disorders (phaeochromocytoma, acromegaly, Cushing syndrome). • Pancreatic disease (pancreatitis), hemochromatosis.	

Type of primary diabetes	Causes	Risk factors
Type 1 {insulin-dependent (IDDM) or juvenile-onset diabetes}. • 5-10% • Most often in children and adolescents • Latent autoimmune diabetes in adults (LADA): It is slow presentation of type 1 diabetes.	Absolute insulin deficiency due to Beta-cell destruction	• Genetic: HLA-DR3/4 in 95%. • Viral (Coxsackie or rubella). • Immune mediated: Associated with autoimmune diseases. Antibodies against insulin and pancreatic islets of Langerhans.
Type 2 diabetes {non-insulin-dependent (NIDDM) or maturity-onset diabetes}. • 90-95% • Maturity-onset diabetes of the young (MODY): • Rare form; it caused by autosomal dominant mutations. • It varies from mild glucose intolerance to insulin-requiring diabetes. • Diagnosed before the age of 25 years.	Predominantly insulin resistance with relative (less) insulin deficiency.	• Family history • Over age of 40. • Overweight, Obese. • Alcohol.
Type 3 diabetes (T3DM) or Gestational diabetes. • Rare type, appears usually in second or third trimester of pregnancy	Affects up to 5% of pregnant women. Placental hormones interfere with insulin	Most common in African. Family history of diabetes. Increased risk of type 2 DM.

Characteristic	Type 1 (10%)	Type 2
Onset (Age)	Usually < 30	Usually > 40
Type of onset	Abrupt	Gradual
Nutritional status	Usually thin	Usually obese
Clinical symptoms	Polydipsia, polyphagia, polyurea, Wt loss	Often asymptomatic
Ketosis	Frequent	Usually absent
Endogenous insulin	Absent	Present, but relatively ineffective
Related lipid abnormalities	Hypercholesterolemia frequent, all lipid fractions elevated in ketosis	Cholesterol & triglycerides often elevated; carbohydrate-induced hypertriglyceridemia common
Insulin therapy	Required	Required in only 20 - 30% of patients
Hypoglycemic drugs	Should not be used	Clinically indicated
Diet	Mandatory with insulin	Mandatory with or without drug

Clinical Presentations

- The majority of patients are asymptomatic.
- Classic symptoms of hyperglycemia (according to blood glucose level):
 - **Polyuria (Excessive urination)**: when serum glucose rises above renal threshold {180 mg/dL} (10 mmol/L), for glucose reabsorption
 - **Polydipsia** (Excessive thirst).
 - **Polyphagia** (excessive eating).
 - **Blurred vision**.
 - Weight loss, fatigue and weakness.
 - **Frequent infections** (genital itchiness due to Candida infection).
 - Slow healing of wounds or sores.

Complications of Diabetes

- Acute complication:

- Diabetic ketoacidosis (DKA):

- It is a medical emergency and may be the initial presentation in 25% of adult newly diagnosed type 1, but may also occur in type 2 (severe infection or acute illness).

- Clinical pictures:

- Disturbed conscious level: lethargy, Confusion, irritability /coma
 - Gastrointestinal: Nausea, vomiting and abdominal pain
 - Smell of acetone in the breath

– **Hyperosmolar non-ketotic acidosis** (rare in type 2 DM)

is another emergency characterized by:

- Severe dehydration secondary to marked **hyperglycemia** {over 600 mg/dL (33.3 mmol/L)}, with **hyponatremia** leading to an altered mental state Drowsiness, Confusion and possibly coma.

– **Lactic acidosis.**

– Hypoglycemia:

- Levels that are ≤ 70 mg/dL
- Causes:
 - Skipping a meal and getting more physical activity than normal.
 - Increase insulin or glucose-lowering

Level 1	Glucose <70 mg/dL (3.9 mmol/L) and ≥ 54 mg/dL (3.0 mmol/L)
Level 2	Glucose <54 mg/dL (3.0 mmol/L)
Level 3	A severe event characterized by altered mental and/or physical status requiring assistance for treatment of hypoglycemia



LOW BLOOD SUGAR



(WITH DIABETES)

- TOO MUCH INSULIN
- TOO MUCH DIABETES MEDICATION

CAUSED BY:

(WITHOUT DIABETES)

- EATING LESS or EXERCISING MORE THAN USUAL
- ALCOHOL
- UNDERLYING CONDITIONS
- SOME MEDICATIONS
- REACTIVE



SYMPTOMS:

- HUNGER
- FATIGUE
- SHAKING
- SWEATING
- PALE SKIN
- HEADACHE
- DIZZINESS



- CONFUSION
- SLURRED SPEECH
- BLURRED VISION
- FAINTING
- SEIZURES
- COMA



15-15 RULE:

1 TEST BLOOD GLUCOSE



2 Eat/Drink 15 GRAMS of FAST-ACTING CARBS



3 WAIT 15 MINUTES



THEN TEST AGAIN

* REPEAT UNTIL BLOOD SUGAR LEVELS ARE > 70 mg/dL

GLUCAGON INJECTION:



INJECT AT 90° ANGLE into THIGH or BUTTOCK



Treatment of Hypoglycaemia

Take quickly absorbed carbohydrates such as:

- half a glass of juice (a little carton or small bottle of fruit juice)
- 5-7 jellybeans
- 3 teaspoons of honey or sugar
- glucose that contains 15 grams of carbohydrate (see package labels)



THEN follow up with more slowly absorbed carbohydrate such as:

- a sandwich
- biscuits
- a glass of milk
- a piece of fruit



Re-test the blood glucose level after 15 minutes



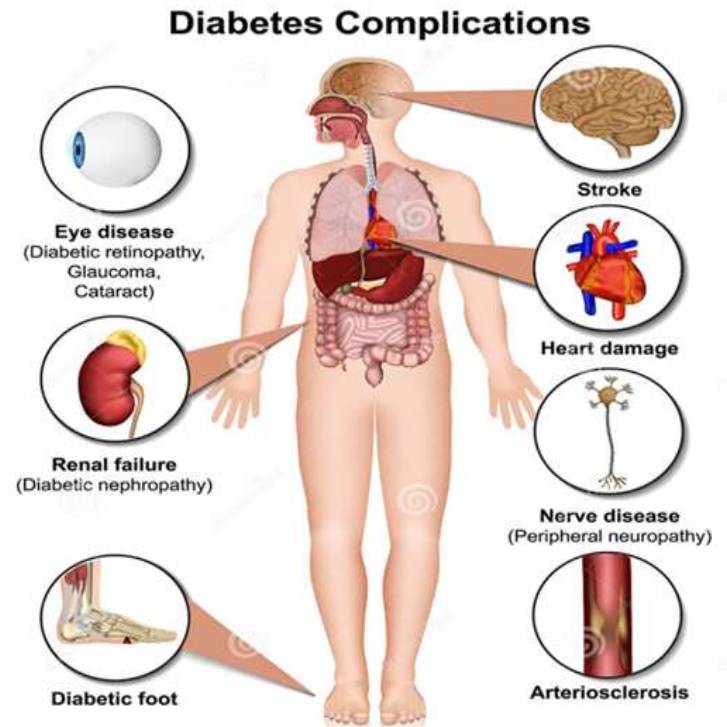
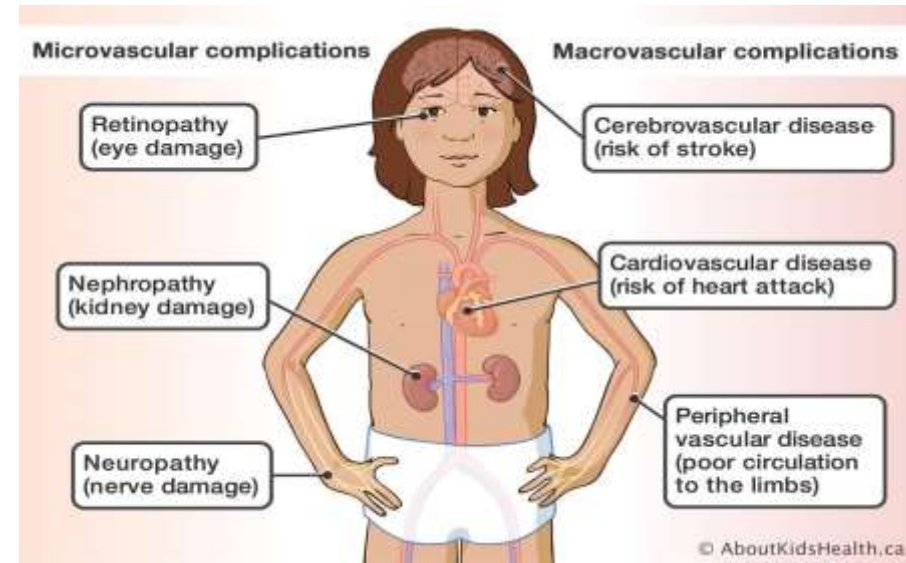
- **Chronic complication:**

- **Micro-vascular:**

- Neuropathy.
 - Retinopathy.
 - Nephropathy.

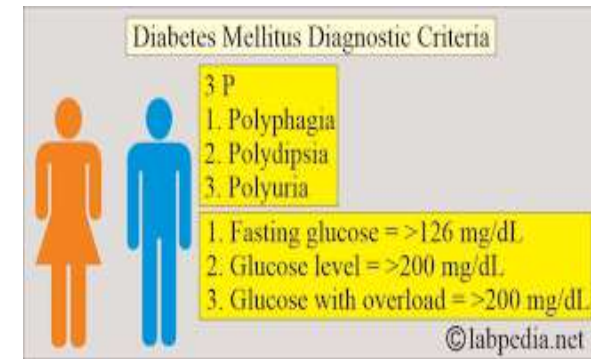
- **Macro-vascular:**

- **Cardiovascular:**
 - Peripheral vascular disease
 - Atherosclerosis (coronary artery disease)
 - Cerebrovascular stroke (CVS)
 - **Renal failure**





Diagnosis



- **History and examination**
- **Blood glucose level (strip test or blood):**
 - **Confirming the diagnosis by repeat measurement on a subsequent day.**
 - **Fasting plasma glucose or sugar (FPG):** fasting 8-12 hours.
 - **Postprandial blood glucose level (PPBG):** 2 hours after meal.
 - **Oral glucose tolerance test (OGTT):** {not commonly used clinically except in pregnancy}
 - **Patient has FBG test, then drink a 75 gram glucose drink (eat). Another blood sample 2 hours after glucose drink.**
 - **Impaired when FBG 100-125 mg/dl and 144-199 mg/dl 2 hours PPBG.**

– **Glycosylated hemoglobin (HbA1c):** reflect BG concentration over the preceding 2-3 months.

- **Normal:** 4- 5.6%
- **Impaired glucose tolerance:** 5.7 -6.4
- **Diabetes:** $\geq 6.5\%$

Normal adult blood sugar level

Types of test	Normal adult blood sugar level
Simple blood sugar level test	80-100 mg/dl
Fasting blood sugar level test	80-100 mg/dl
Oral-Glucose-Tolerance-Test	<140 mg/dl
A1C Test	4% - 6%

- **Others:**
 - **Kidney function test and electrolytes.**
 - **Lipid profile.**
 - **Arterial blood gas (ABG).**
 - **Urine albumin-to-creatinine ratio and urine analysis.**

American Diabetes Association (ADA) criteria for diagnosis of DM

- **A1C $\geq 6.5\%$.**
- **Fasting plasma glucose (FPG) level ≥ 126 mg/dL (7.0 mmol/L), or**
- **2-hour plasma glucose level ≥ 200 mg/dL (11.1 mmol/L) during a 75-g oral glucose tolerance test (OGTT), or**
- **Random plasma glucose ≥ 200 mg/dL (11.1 mmol/L) in a patient with classic symptoms of hyperglycemia or hyperglycemic crisis**

	No Diabetes	Prediabetes	Type 2 Diabetes
HbA1c	4 - 5.6%	5.7 - 6.4%	6.5+%
Fasting Blood Glucose	<100 mg/dL	100-125 mg/dL	126+ mg/dL
OGTT	< 140mg/dL	141-199 mg/dL	200+ mg/dL

- **ADA recommends that the following people be screened for diabetes:**
 - **Anyone with a body mass index higher than 25** (23 for Asian Americans), regardless of age, **who has additional risk factors** {high blood pressure and cholesterol levels, inactive lifestyle, a history of polycystic ovary syndrome or heart disease, and family history of diabetes}.
 - **Anyone older than age 35** is advised to get an initial blood sugar screening. If the results are normal, they **should be screened/ three years** after that.
 - **Women who have had gestational diabetes** are advised to be screened/ three years.
 - **Anyone who has been diagnosed with prediabetes** is advised to be tested every year.
 - Anyone who has HIV is advised to be tested.

Treatment

- **Type 1 DM:** require lifelong insulin therapy.
 - Many types of insulin, including short-acting (regular insulin), rapid-acting insulin, long-acting insulin and intermediate.
 - Most require 2 or more injections/daily, with doses adjusted on self-monitoring of blood glucose levels.
 - **Continuous subcutaneous insulin infusion (CSII):**
 - Rapid-acting insulin infused continuously 24 hours a day through an insulin pump.
 - **Side effect of insulin:** hypoglycemia
 - **Other:**
 - Pancreatic transplantation or islet cell transplant.



- **Type 2 diabetes:**

- **Life style modification**

- **Diet**

- **Recommendations for amounts of dietary carbohydrate, fat, and protein.**

- **Instructions on how to divide calories between meals and snacks.**

- **More fruits, vegetables, lean proteins and whole grains.**

- **Activity or exercise:** at least 30 minutes or more of moderate physical activity most days of the week

- **Anti-diabetic medications or Oral hypoglycemic drugs (metformin).**

Class	Agent	Method of Action	Risk of Hypoglycemia	Side Effects
Biguanides	Metformin	Reduce insulin resistance in liver	Mild	Gastrointestinal; bloating, nausea, hyperdefecation, diarrhea
Thiazolidinediones	Rosiglitazone Pioglitazone	Reduce insulin resistance in fat and muscle	Mild	Weight gain, fluid retention
α -glucosidase inhibitors	Acarbose	Impairs carbohydrate absorption	None when used as monotherapy	Gastrointestinal: bloating, flatulence
DPP4 inhibitor	Sitagliptin	Prolongs action of endogenous incretins	None when used as monotherapy	Headache, runny or stuffy nose
Sulfonylureas	Chlorpropamide Glipizide Glyburide Glimepiride	Bind to receptor in islet cell; insulin secretagogue	Significant	Weight gain, allergic sensitivity in those with sulfa allergy
Meglitinides	Repaglinide Nateglinide	Increase synthesis of insulin; insulinotropic	Moderate	Back pain, headache, diarrhea

- **Insulin**
- **Goal of treatment:**

Pre-meal (Fasting): 80-120 mg/dL

Post meal 2 hrs (postprandial): 80-140 mg/dL

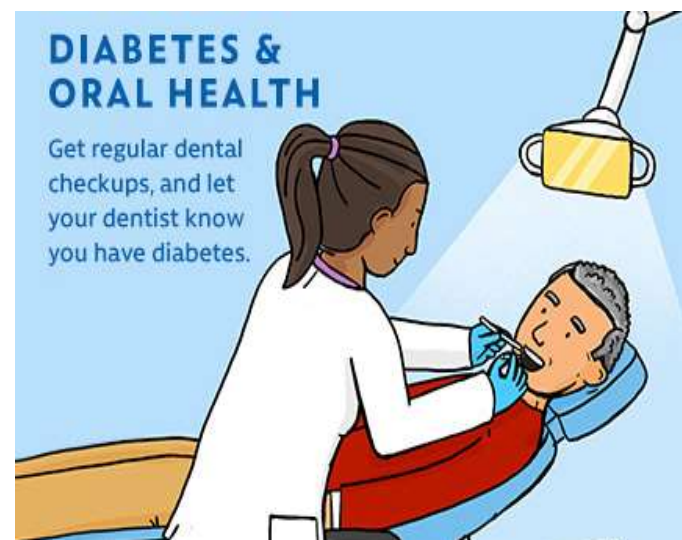
HbA1c: American diabetes association (ADA) goal is $\leq 7\%$,
American Association of Clinical Endocrinologists (AACE)
goal is $\leq 6.5\%$.

Treatment of Complications

Hypoglycaemic coma	Hyperglycaemic coma
Diabetes usually known	Diabetes may be unrecognized
Too much insulin or too little food, or too much exercise or alcohol	Too little insulin, infection, or myocardial infarct
Adrenaline release causes: Sweaty warm skin Rapid bounding pulse Dilated (reacting) pupils Anxiety, tremor, aggression Tingling around mouth	Acidosis in some variants causes: Vomiting Hyperventilation Ketonuria Acetone breath
Cerebral hypoglycaemia causes: Confusion, disorientation, aggression Headache Dysarthria Unconsciousness Focal neurological signs (e.g. fits)	Osmotic diuresis and polyuria cause: Dehydration Hypotension Tachycardia Dry mouth and skin Abdominal pain
Management	
Take blood for baseline glucose, electrolytes, urea, Hb and PCV	
If conscious, give Hypostop (GlucoGel) or 25g glucose orally (2 teaspoons sugar, 3 lumps sugar, 3 Dextrosol tablets, 60ml Lucozade, 15ml Ribena, 90ml cola drink (not 'diet' variety) or 1/3 pt milk)	Put up infusion to rehydrate and give insulin
If comatose, give 20mg 20% or 50% dextrose i.v. and on arousal 25mg orally	
If i.v. injection impossible, give glucagon 1mg i.m.	
Call ambulance (Ch. 1)	

Schedule For Routine Medical Care

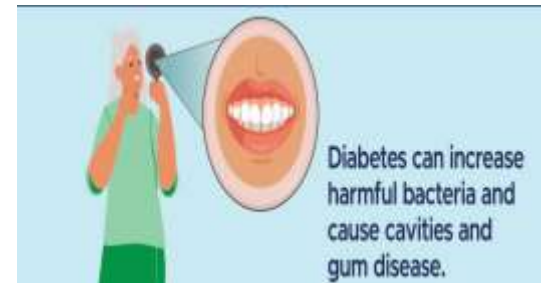
HbA1c	2- 4 times/year
Blood pressure	At least 2 times/year
Cholesterol	At least every other year
Dilated eye exam	1 time/year
Foot exam	At least 1 time/year
Dental exam	2 times/year
Urine microalbumin/ creatinine ratio	1 time/year
Flu shot	1 time/year
Pneumococcal vaccine	Once in life (repeat at age 65)



Dental considerations of DM or

Effect of DM, its complications and treatment on dental health

- Diabetes increases the risk of infection and also slows healing, so any dental surgery will take longer to heal.
- There are **no specific oral manifestations of DM**, but uncontrolled diabetic patients (22%) are **predisposed to periodontal disease gingivitis** (swollen, tender, or bleeding gums), if not treated leading to **severe periodontitis** causing:
 - Pain, Persistent bad breath (halitosis)
 - Chewing difficulties, Tooth loss If not treated.
 - **Treatment:** scaling and root planing,



- **Xerostomia {dry mouth}** lead to soreness, ulcer, infection and tooth decay (dental caries)
 - **Treatment:** **Avoid dry mouth by** drink plenty of water and chew sugar-free gum to stimulate saliva flow.
- **Fungal infection (thrush)** causes painful ulcerated white patches lead to bad breath and a burning sensation in your mouth, and change your ability to taste.

– **Treatment:**

- Good mouth hygiene
- Control blood glucose levels can help to treat oral thrush.
- **Antifungal** medications if needed.



- **Mucormycosis:** rare, invasive fungal infection in patient with uncontrolled diabetes {especially DKA}.
- **Severe dento-alveolar abscesses** should be investigated to exclude other immune defects.
- **Impaired taste sensation (hypogeusia)** influenced by peripheral neuropathy.
- **Swelling of salivary glands (sialosis)** due to autonomic neuropathy.

- **Autonomic neuropathy** can cause orthostatic hypotension; **therefore the supine patient** should be **slowly raised** upright in the dental chair.
- **Circumoral paraesthesia** (numbness/tingling around the **mouth**) is a critical, often early symptom of impending hypoglycemia in insulin-treated patients.
- **Temporary lingual and labial paraesthesia/burning mouth sensation** may follow the **removal of mandibular third molar teeth**.

- **Grinspan syndrome:**

- Includes diabetes, lichen planus and hypertension.

- **Symptoms:**

- Oral lesions found on the buccal mucosa, tongue, and gums, presenting as white streaks, papules, or ulcers that may cause burning sensations.
 - May be due to drug use.



- **Drugs:**

- Avoiding drug can disturb diabetic control:
 - **Corticosteroids**, which **increase blood glucose**.
 - **Doxycycline, tetracycline and ciprofloxacin**, enhance **insulin hypoglycemia**.
- Paracetamol (acetaminophen) or codeine is the **analgesic of choice**.
- **NSAIDs** should be used with **caution in renal damage** and risk of **gastrointestinal bleeding**, especially in diabetics are already on low-dose **aspirin** (ischemic heart disease).

- **Amoxicillin is the antibiotic of choice** for treatment of infections (infection aggravate hyperglycemia).
- **LA containing adrenaline safely** used with no special precaution in non-surgical or short minor surgical procedures (simple single extractions).
- **CS with benzodiazepines** can usually be safely used.

Tips for maintaining oral and dental health in a diabetic patient

- **Good oral health habits:** prevent infections.
 - Brushing (soft-bristled brush) with fluoride toothpaste after meals or at least twice a day.
 - Teeth floss at least once/day or interdental cleaners.
 - Maintain regular dental Visit {once or twice/year} to detect any signs of oral infection {morning appointments are advisable}.
 - Eat a healthy diet and get plenty of physical activity.
 - Smoking cessation as smoking increases the risk of periodontal disease and can worsen diabetes.
 - If you have dentures or removable bridgework, clean them every day.

Dental surgery or operation and DM

- **Early preoperative assessment:**
 - **High-risk patients** (poor glycemic control/complications of diabetes).
 - **Kidney function test and electrolytes.**
 - **Perioperative diabetes management strategy (use insulin).**
 - The physician may need to adjust the patient's diabetes medication to ensure sustained metabolic control, before, during, and after surgical procedures.
 - **Identify and treat other co-morbidities** (cardiac, renal and neurological)

- **Starvation time should be minimized** (hypoglycemia should be avoided).

Also **hyperglycemia** may occur due to stress or trauma.

- **Prevention and treatment:**

- **Monitor blood glucose** (fingerpick test) **before, during** (hourly) and **after** surgery (immediately and regularly thereafter).
- Oral glucose just before the appointment if blood glucose level is too low or a patient has taken their medication without appropriate meal.
- Postoperative plasma glucose level and further glucose can be given.

- **Elective dental treatment** may be delayed until the patient's diabetes is considered stable or better controlled.
- **Surgery perform** in early morning with no food & no drugs.
 - Target blood glucose should be 6–10 mmol/L (108-180 mg/dL) (acceptable range 4–12 mmol/L).

- **Minor surgical procedures (simple single extractions) under LA:**

- **Well-controlled diabetic patient**, procedure is carried out within 2 hours of breakfast and morning insulin injection, with **no change in insulin regimen**.
- **Poorly controlled diabetics** (type 1 or 2 DM) should also be referred for **control their blood glucose before non-emergency treatment**.

- **Minor operations (multiple extractions or endoscopic procedures) and/or major operation under GA in hospital early morning withholding both food and insulin from the previous midnight until finished.**
- **Patients undergoing major operations will require insulin therapy preoperatively and stabilized over 48 hours.**
 - **Patients who are inadequately controlled on oral therapy (random BG >15.0 mmol/L or HbA1c >9%).**

- **Postoperatively:**

- **Return to normal diet** and usual diabetes regimen (insulin or oral hypoglycemic).
- **Oral hypoglycemic drugs except metformin**, should be **restarted** as normal with the first meal.
 - **Metformin** should not be used if the estimated glomerular filtration rate (eGFR) less than 30 mL/min/1.73 m²

Management of diabetes requiring general anesthesia

	Type 1 DM	Type 2 DM
Preoperative	Stabilize on at least twice-daily insulin for 2–3 days preoperatively. One day preoperatively, use only short-acting insulin (Actrapid soluble or neutral).	Stop biguanides.
Perioperative	Do not give sulphonylurea or SC insulin on day of operation. Estimate blood glucose level. Set up intravenous infusion of 10% glucose 500 mL containing Actrapid or neutral insulin 10 units plus potassium chloride 1 g at 8.00 am. Infuse over 4 h. Estimate blood glucose and potassium levels 2-hourly. Adjust insulin and potassium to keep glucose at 5–10 mmol/L and normokalaemic	Omit oral hypoglycemic drugs. Estimate blood glucose level.
Postoperative	• Continue infusion 4-hourly. • Estimate blood glucose 4-hourly. • Estimate potassium 8-hourly	Estimate blood glucose 4 h postoperatively.
On resuming normal diet	Stop infusion. Start Actrapid or neutral insulin and continue over the next 2 days. Start sulphonylurea. Start normal insulin regimen	Start sulphonylurea or other usual regimen.

- *People with diabetes often think about possible long-term complications such as heart disease and nerve damage. But they shouldn't forget about what's going on in their mouth. Good dental care and diabetes care have a direct impact on their future ability to chew and smile.*

Thank You



“Keeping your blood sugar levels near target—by eating a healthy diet, taking your medications as prescribed, and being active—is the key to good oral health”.